

DUSTRUNNER · DESIGN PRECIS

DustRunner design precis

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3 update. Design choices decided per DRN-DDR-001; numbers replaced by DRN-CAL-001 (50 mm brush core, 70 N hook preload, 0.8 mm hood, dock anemometer); model, drawing DRN-DWG-001 and media refreshed
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, traction, wind, economics, safety, media)
0.1	2026-09-24	Amish Chadha	Initial scaffold

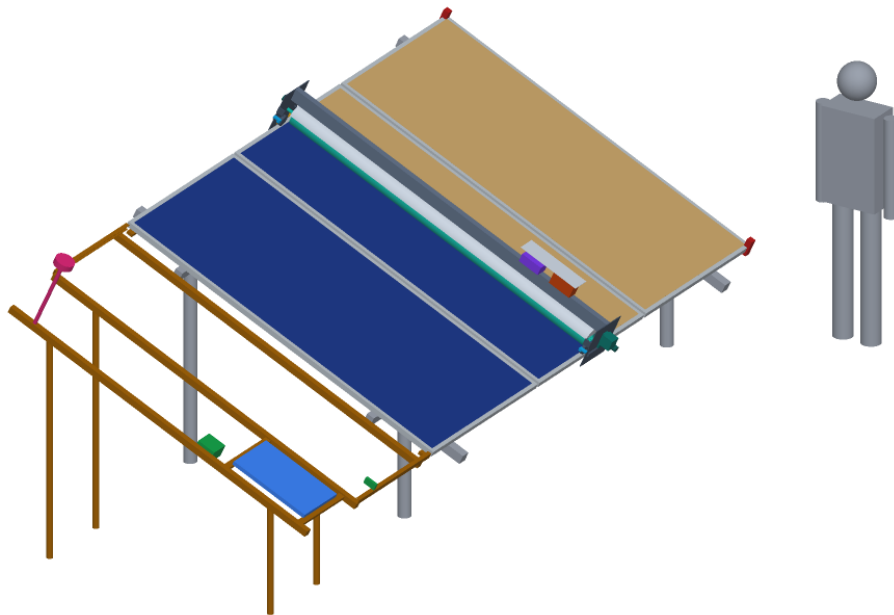
DustRunner design precis

■ Summary

DustRunner is a rail-free crawler robot that spans one PV table from its lower to its upper module edge, rides along the module frames on two end trucks that hook under the frame lips, and sweeps the glass dry with a 2.2 m rotating microfiber brush. It parks off the glass in a clamp-on dock at the start of the row, where a 20 W panel recharges its 12.8 V LiFePO4 pack and a cup anemometer tells it whether the wind allows a run. The TRL 3 calculations (DRN-CAL-001) give, for the 40 m, 20 kWp reference row, about 7.4 minutes and 7.6 Wh per out-and-back cycle, a robot of about 13.8 kg and \$500 in parts for robot, dock and anemometer. Six requirements are met on paper and none is clearly missed, but five are at risk: 5 mm frame steps (R4), traction in an 8 m/s wind (R9), a brief 72 N wheel load at row entry (R10), pack temperature (R11) and a budget with no margin (R13). Payback is 2.8 years on a 60 m row, which meets R14 as restated, and 4.1 years on the 40 m reference row.

DustRunner

CONCEPT, NOT FOR FABRICATION



Grey: 1.75 m person for scale

Figure 1. DustRunner partway along a three-module section of the reference table, moving away from its dock (brown, left, with the anemometer mast) toward the end stops (red). Glass behind the robot is clean; tan shows dust still to be removed. Grey figure: 1.75 m person for scale.

■ How it works

1. **Park and charge.** Between cycles the robot sits in the dock, a short pair of rails that continue the module frame profile past the end of the row. Parking off the glass means the robot never shades the array. A 20 W panel in the dock charges the pack through spring contacts; a third contact carries the anemometer signal.
2. **Start.** Once a day, in the evening after the array's output has dropped and before dew forms (DRN-DDR-001, D2), the controller checks pack charge, air humidity and the wind at the dock anemometer, sounds a short warning, then drives out of the dock.
3. **Travel.** Two end trucks carry the robot along the row. Each truck has two 70 mm polyurethane wheels on the top flange of the module frame, two guide rollers against the outer face of the frame and a hook roller under the frame's bottom flange, pulled up by springs to 70 N. The trucks are the "clamp": they cannot lift off or slide down the slope, yet need no rail. A gearmotor with an encoder on each truck drives both of that truck's wheels through a toothed belt, and the controller keeps the two ends in step.
4. **Clean.** A 120 mm microfiber brush on a 50 mm aluminum core spans the slope and turns at about 150 rpm, pressed 4.5 mm into the glass plane at its bearings. A helical pattern on the brush conveys the loosened dust down the slope and off the lower edge. A hood over the brush limits dust thrown back onto clean glass.
5. **Stop and return.** IR sensors on each truck see the frame end; a clamp-on end stop at the far end is the mechanical backstop. The robot reverses the brush and cleans on the way back, then docks and latches.

DustRunner: energy per cleaning cycle, 40 m row out and back, Wh (estimates, DRN-CAL-001 F7)

CONCEPT, NOT FOR FABRICATION

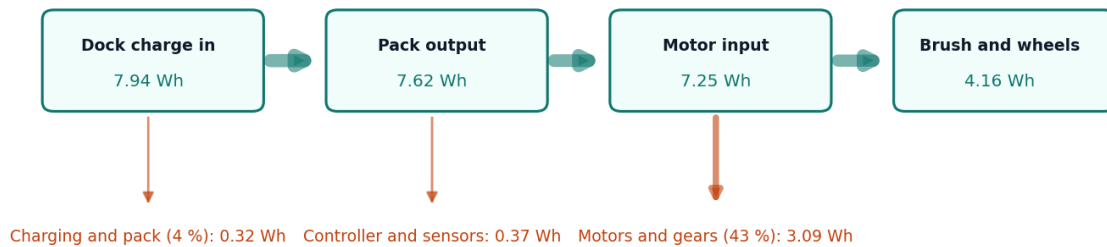


Figure 2. Energy for one out-and-back cycle on the 40 m reference row, in Wh (DRN-CAL-001, F7). All values are estimates.

■ Main components

Numbers match the exploded view (Figure 3) and `bom/bom.csv`. Geometry is in `cad/src/model.py` and drawing DRN-DWG-001.

Table 1. Main components.

#	Component	Choice	Notes
1	Chassis beam	40 x 80 x 2 mm 6063 aluminum tube, 2,322 mm	Cut to the module length for other tables
2	Microfiber brush	120 mm diameter, 2,198 mm brushed length, on a 50 x 2 mm aluminum core	The 50 mm core keeps the interference at 3.4 to 4.4 mm along the brush
3	Brush hood	0.8 mm aluminum sheet, 77 mm radius, with end guards and a sunshade over the pack	Also the finger guard at the brush ends
4	Brush motor and belt drive	12 V DC gearmotor, about 60 W, toothed belt	Reversible; about 150 rpm at the brush; about 47 W in use
5	End trucks and wheels	5 mm aluminum plates, 260 x 272 mm; two 70 x 18 mm polyurethane wheels each at 180 mm wheelbase, belt-linked	Wheels run on the frame top flange only
6	Drive gearmotors	Two 12 V 37 mm gearmotors with encoders, one per truck	Synchronized in firmware
7	Guide and hook rollers	22 mm guide rollers on the frame face; 16 mm hook rollers under the frame bottom flange, 70 N spring preload per truck	Keep the robot on the table in wind
8	Battery	LiFePO ₄ , 12.8 V, 10 Ah (128 Wh) with BMS and temperature cut-offs	Under the sunshade
9	Controller	ESP32-class board, two drive channels, brush driver, current sensing, humidity and temperature sensor, IP65 box	Firmware is a sketch until TRL 4
10	IR edge sensors	Four downward-looking reflective sensors, two per truck	Row-end and gap detection
11	Dock frame	Aluminum C-section rails continuing the module frame profile, cross members, two clamps to the first module frame and four legs	Clamp-on; no drilling
12	Dock PV panel	20 W, about 530 x 350 mm, in the table plane	Charges the parked robot
13	Dock charger and contacts	LiFePO ₄ charge controller, three spring contacts, latch with a 6 mm pin	
14	End stops	Pair of clamp-on blocks with rubber buffers on the last module's frame	Mechanical backstop
16	Dock anemometer	Cup anemometer with pulse output on a 450 mm mast	Read before each start (DRN-DDR-001, D3)

Item 15 (wiring, fuse, switches, stop buttons and fasteners) has no callout.

DustRunner: exploded view (robot laid flat; dock and end stop drawn beside it)

CONCEPT, NOT FOR FABRICATION

- 1 Chassis beam, 40 x 80 mm
- 2 Microfiber brush, 120 mm x 2.2 m
- 3 Brush hood
- 4 Brush motor and belt drive
- 5 End trucks and wheels (pair)
- 6 Drive gearmotors (2)
- 7 Guide and hook rollers (set)
- 8 LiFePO4 pack, 12.8 V 10 Ah
- 9 Controller and motor drivers
- 10 IR edge sensors (4)
- 11 Dock frame, rails and clamps
- 12 Dock PV panel, 20 W
- 13 Dock charger and contacts
- 14 End stops (pair)
- 16 Dock anemometer

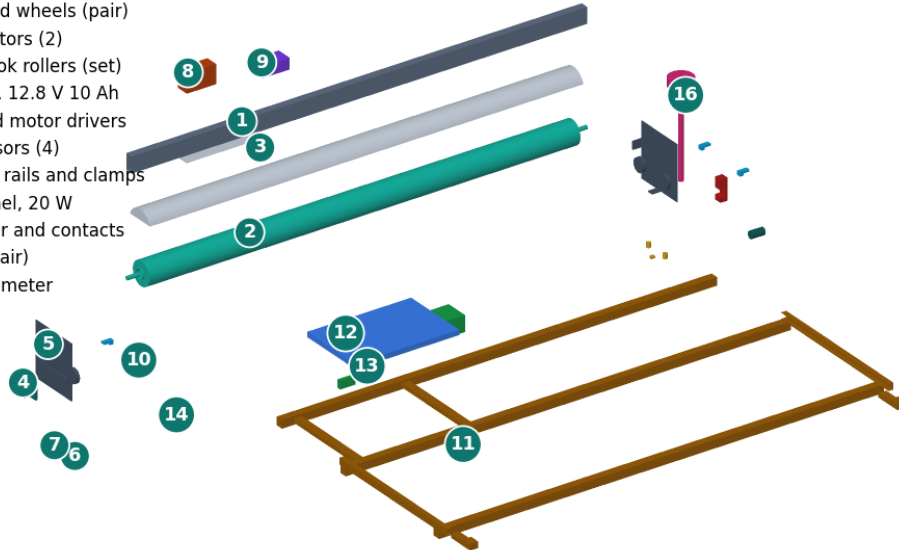
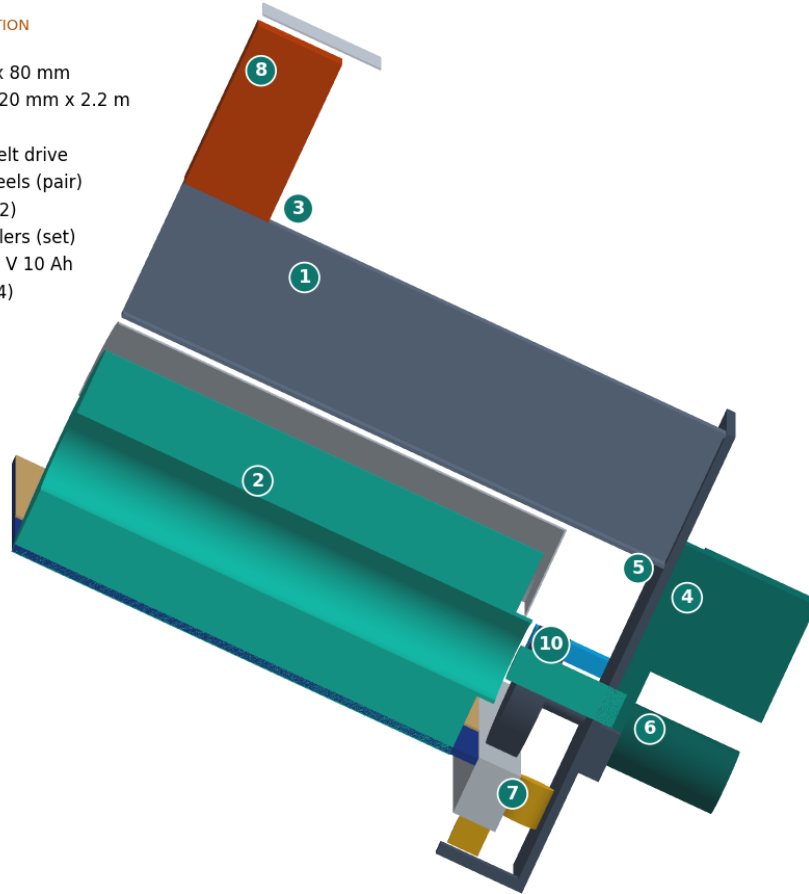


Figure 3. Exploded view with numbered callouts matching the BOM. The robot is laid flat in the table plane; the dock and end stops are drawn beside it. Upper-end twins of paired parts are shown without numbers.

DustRunner: section across the row at the lower module edge

CONCEPT, NOT FOR FABRICATION

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- 8 LiFePO4 pack, 12.8 V 10 Ah
- 10 IR edge sensors (4)



Silver: module frame; blue: glass. Truck (5) wheels ride the frame top; rollers (7) bear on the frame face and hook under its lip

Figure 4. Section across the row through the robot at the lower module edge, showing how the end truck wraps the frame.

■ Numbers at TRL 3

All values are from DRN-CAL-001 v0.1, which prints each from `docs/04-calcs/sizing.py`. They are first-principles estimates; the assumptions behind them (pile pressure, friction coefficients, frame flange width) are listed there and must be measured.

Table 2. Key numbers.

Quantity	Value	Requirement
Robot mass	13.8 kg	R10 met (15 kg)
Brush contact	About 67 N on the glass, 30 N drag, 28 W mechanical (13 to 50 W over the assumed ranges)	
Electrical power while cleaning	67 W (brush 47 W, drives 17 W, electronics 3 W)	
Wheel load	55 N brushing at 25°; 58 N at worst tilt; about 72 N for about a second at row entry	R10 at risk
Traction margin, friction 0.4	1.84 in still air; 1.10 in an 8 m/s wind along the row	R9 at risk
5 mm frame step	About 9 N extra push needed, about 14 N spare	R4 at risk
Parked at 35 m/s	555 N along the row on a latch rated about 2.8 kN; 137 N net lift on the hook rollers	R9 parked case met
40 m row, out and back	7.4 min; 7.6 Wh from the pack	
100 m row, out and back	17.4 min; 19.6 Wh from the dock	R7 met
Autonomy	Three 100 m cycles and three days of standby use 67 of 102 Wh; 9 days with no sun on the 40 m row	R8 met
Dock harvest	45 Wh/day at 3 sun hours; 82 Wh/day at 5.5	R8 met
Coverage	98.4 % of the glass on the 40 m row	R5 met
Stopping	Brush 0.13 s, robot 0.06 s after power is cut	R12 met
Parts cost	\$500.00 (16 lines)	R13 at risk, no margin
Payback, mid case	2.8 years on a 60 m row; 1.6 years on 100 m; 4.1 years on the 40 m reference row	R14 met as restated

The pack is cycled lightly (under 10 % of capacity a day on the reference row), which is kind to LiFePO4 cells in heat. The beam is not critical: with its load and a 250 N point load at mid-span it is stressed to about 17 MPa and deflects about 2.9 mm. Water saved is about 122 to 350 L per avoided wash of the reference row, about 1.5 to 4.2 m³ a year for monthly washing.

■ Key design choices

Amish decided the design choices below on 2026-09-25 by approving the TRL 2 recommendations (DRN-DDR-001). Choices raised by the TRL 3 calculations are listed after them as proposals.

- **One robot per row, spanning the full slope (D1).** A full-width robot needs no steering on the glass and cannot fall between modules. The concept is limited to tables whose short edges are free of clamps, as stated in R4.
- **Ride on the frame edges, not on the glass (D1, D10).** Wheels on the frame keep weight off the glass and coating. The first supported format is 1P portrait tables of 2,278 mm modules.

- **Rail-free, with a short dock and clamp-on end stops** (D11). The dock and stops are the only added hardware and clamp on without drilling. The mechanical end stops stay as the independent backstop to the IR sensors.
- **Dry microfiber brush, no water and no airflow** (D12). Airflow is added only if tests show that dust is re-deposited.
- **Charging at the dock from a 20 W panel** (D4), not from a panel on the robot.
- **Both wheels driven on each truck, with preloaded hook rollers** (D5). DRN-CAL-001 sets the preload at 70 N per truck.
- **Evening cleaning, gated by the humidity sensor** (D2). While on the glass the robot shades a strip along one module for a few seconds, which costs 1 to 2 kWh a year.
- **Anemometer at the dock** (D3), read through the dock contacts before each start.
- **LiFePO4 at 12.8 V, 10 Ah** (D6). A SwapCell pack is not used: at about 468 Wh and 48 V it is far larger than DustRunner needs.
- **Reference row and payback target** (D7, D8). The 40 m reference row is kept; R14 applies to rows of 60 m or more.
- **Budget** (D9). `budget_usd` stays at \$500 for robot, dock and anemometer. The priced BOM uses all of it.

Proposed, awaiting Amish (from DRN-CAL-001; details in `docs/REVIEW.md`): a lower wind limit for starting a run, how to treat the 72 N row-entry wheel load, and whether to hold a cost contingency.

■ Safety

Safety: DustRunner is moving machinery with a rotating brush, a lithium iron phosphate battery, and it works on an energized PV array that can carry several hundred volts DC, on hot glass, up to about 1.6 m above the ground.

- **Rotating brush and pinch points.** The brush, belts and wheels can catch fingers, hair and clothing. The hood covers the brush ends; each truck has an emergency stop button; the brush stops when a truck lifts off the frame. The brush coasts to rest in about 0.13 s once power is cut (DRN-CAL-001, H1). Nobody should reach under the hood while the robot is powered.
- **Unexpected start.** The robot runs on a timer with nobody present. It must carry a clear warning label, start only after a short audible warning, and have a lockout switch at the dock.
- **Falling robot.** A 13.8 kg robot leaving the row end could injure someone below. Two independent stops (IR sensing and the mechanical end stop) and the hook rollers address this (R9). The end stop and dock latch hold with factors above 4 on paper, but traction in wind is marginal, so the start wind limit matters (DRN-CAL-001, C4, D1 to D4).
- **PV array electrical hazard.** Strings carry DC voltages that can be lethal and cannot be switched off in daylight. Nothing on the robot may touch cables, connectors or junction boxes; installation and maintenance follow the array owner's electrical safety rules. Damaged module glass exposes live parts.
- **Battery.** LiFePO4 is less prone to thermal runaway than other lithium-ion chemistries but can still vent and burn if crushed, shorted or overcharged. Use a pack with a BMS, fuse the pack output, block charging below 0 °C and above about 45 °C, and keep the pack shaded and ventilated (R11 is at risk on pack temperature).
- **Hot surfaces.** Module glass and frames can reach about 75 °C. Wear gloves when installing or servicing in daylight.

- **Module warranty.** Abrasive cleaning, or a cleaning tool the maker has not cleared, can void the module warranty. Check the maker's cleaning guidance and ask the maker to clear the robot before use (see DRN-PRB-001).

■ Open questions

- [] Brush pile pressure, drag power and cleaning effectiveness on real dust, dry and after dew (R2).
- [] Long-term abrasion of anti-reflective coatings by daily microfiber contact (R3).
- [] Friction coefficient of polyurethane on dusty frames (R9); the traction margin in wind depends on it.
- [] Visible frame flange width of target modules; the wheel tread needs about 21 mm (R4).
- [] Pack temperature when parked in the sun, and the charging window it allows (R11).
- [] Module makers' positions on frame loading, including the 72 N row-entry transient, and on robotic dry cleaning (R10).
- [] How dust conveyed off the lower edge builds up at the lower frame lip over time.
- [] Pilot site and co-design partner (DRN-DDR-001, O1).